CBCS SCHEME

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BPLCKC205/BPLCK205C

Second Semester B.E./B.Tech Degree Examination, Dec.2023/ Jan.2024
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain abstraction and inheritance in detail.	6	L2	CO1
	b.	Write a simple java program and explain compilation process.	6	L3	CO1
	c.	Describe the meaning of each keyword in "Public static void main".	8	L1	CO1
		OR			
Q.2	a.	Write a note on : i) Integer Data types ii) Boolean Data types.	8	L1	CO2
	b.	Define a variable. Explain declaration and initialization of a variable.	6	L1	CO2
	c.	What is an array? Write a program to demonstrate a one-dimensional array.	6	L3	CO2
		Module – 2			
Q.3	a.	Explain bitwise logical operators with an example program.	10	L1	CO2
	b.	Explain : i) Increment and decrement operators ii) Ternary operator.	10	L1	CO2
		OR			
Q.4	a.	Illustrate switch statement with syntax and example program.	10	L3	CO2
	b.	Explain any two iterative statements with an example program.	10	L3	CO2
		Module – 3			
Q.5	a.	What are constructors? Explain parameterized constructor with an example program.	10	L2	CO2
	b.	Explain : i) Instance variable hiding ii) The finalize( ) method.	10	L2	CO3
		OR			
Q.6	a.	What is Recursion? Explain with an explain program.	10	L2	CO3
	b.	Write a Java program to demonstrate method overloading and constructor overloading.	10	L3	CO3
		Module – 4			
Q.7	a.	Explain two uses of super with an example.	10	L1	CO3
	b.	Explain dynamic method dispatch with an example program.	10	L2	CO3
		OR			
Q.8	a.	Define abstract class. Explain with an example program.	10	L3	CO4
	b.	What is method overfeeding? Explain with are example program.	10	L2	CO4
		Module – 5			
Q.9	a.	What is package? Explain with an example program.	10	L1	CO4
	b.	Write a Java program to read two integers a and b. Compute a/b and print when b is not zero. Raise an exception when b is equal to zero. Also demonstrate working of array index out of Bound Exception.	10	L3	CO4
		OR			
Q.10	a.	Define interface. Demonstrate the working of interface in Java with an example.	10	L1	CO4
	b.	Explain chained Exception with an example program.	10	L1	CO4

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BPLCK105C/BPLCKC105

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Basics of Java Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the basic principles of object oriented programming.	6	L1	CO1
	b.	Explain primitive data types used in java.	8	L1	CO1
	c.	What is type conversion and type casting? Explain with an example.	6	L1	CO1
		OR			
Q.2	a.	Explain two control statements used in Java with syntax and example.	6	L1	CO1
	b.	What is multidimensional array? Explain with an example program.	8	L1	CO1
	c.	Explain lexical issues used in java.	6	L1	CO1
		Module – 2			
Q.3	a.	Write a java program to show the use of the following operators: i) << ii) >> iii) >>>	6	L3	CO2
	b.	Explain logical operators with example program.	8	L1	CO2
	c.	Explain for each version of for loop with an example.	6	L3	CO2
		OR			
Q.4	a.	Write a note on: i) Break ii) Continue.	6	L1	CO2
	b.	Explain different types of if statement with example program.	8	L3	CO2
	c.	Write a program to demonstrate the use of ternary operator.	6	L3	CO2
		Module – 3			
Q.5	a.	Explain two ways of argument passing methods with example.	6	L1	CO3
	b.	What are constructors? Explain two different types of constructors with example program.	8	L3	CO3
	c.	Define method overloading. Explain constructor overloading with an example.	6	L2	CO3

		OR			
Q.6	**a.**	Write a program to perform stack operations using proper class and methods.	**6**	**L3**	**CO3**
	b.	What is recursion? Explain with an example program.	**8**	**L1**	**CO3**
	c.	Explain: i) This keyword ii) The finalize( )	**6**	**L1**	**CO3**
		Module – 4			
Q.7	**a.**	What is Inheritance? Explain how a super class variable can reference a subclass variable.	**6**	**L2**	**CO3**
	b.	Explain method overriding with example program.	**8**	**L3**	**CO4**
	c.	Describe the two uses of super keyword with example.	**6**	**L3**	**CO4**
		OR			
Q.8	**a.**	Write a note on: i) Final class ii) Final method.	**6**	**L1**	**CO4**
	b.	Explain dynamic method dispatch with example.	**8**	**L3**	**CO4**
	c.	What is abstract class? Explain with an example program.	**6**	**L3**	**Co4**
		Module – 5			
Q.9	**a.**	Define package. Explain with an example program.	**6**	**L3**	**CO4**
	b.	What is an exception? Explain different mechanisms of handling the exceptions in java.	**8**	**L3**	**CO4**
	c.	Explain nested interfaces with example.	**6**	**L1**	**CO4**
		OR			
Q.10	**a.**	Describe different access specifiers used in java.	**6**	**L1**	**CO4**
	b.	Define Interface. Demonstrate the working of interface.	**8**	**L1**	**CO4**
	c.	Explain how one interface can be extended by another with an example program.	**6**	**L3**	**CO4**

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Model Question Paper-I/II with effect from 2022-23 (CBCS Scheme)

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First/Second Semester B.E. Degree Examination

Introduction to JAVA Programming

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02. Use JAVA code snippet to illustrate a specific code design or a purpose.

		Module -1	Bloom's Taxonomy Level	Marks
Q.01	a	Explain object oriented principles.	L2	6
	b	Describe the meaning of each of the keyword in "public static void main"	L1	6
	c	Explain different lexical issues in JAVA.	L2	8
		OR		
Q.02	a	Explain different promotion rules in JAVA.	L2	6
	b	Give different syntax of array allocation in JAVA.	L1	6
	c	Explain different type of integer literals with examples.	L2	8
		Module-2		
Q. 03	a	Explain the following operations with example. (i) << (ii) >> (iii) >>> (iv) &	L2	8
	b	Demonstrate the working of enhanced for loop.	L2	6
	c	Write a program to sort the elements using for loop.	L3	6
		OR		
Q.04	a	Explain four different types of if statements in JAVA.	L2	8
	b	Demonstrate working of break with labels in JAVA.	L2	6
	c	Write a program to find the biggest of three numbers using ternary operator.	L3	6
		Module-3		
Q. 05	a	What are constructors ? Explain two types of constructors.	L2	6
	b	Explain static variable and static methods in JAVA.	L2	6
	c	Write a program to perform Stack operation using proper class and methods	L3	8
		OR		
Q. 06	a	Explain memory allocation and use of garbage collector in JAVA	L2	6
	b	Explain use of *this* in JAVA.	L2	6
	c	Write a JAVA program demonstrating Method overloading.	L3	8
		Module-4		
Q. 07	a	Explain different types of inheritance in JAVA.	L2	6
	b	Example two uses of super keyword with example.	L2	6
	c	Demonstrate the working of final class and final method in JAVA.	L3	8
		OR		
Q. 08	a	What is abstract class ? Explain with example.	L2	6
	b	Give example for method overriding.	L2	6
	c	Demonstrate dynamic dispatch using abstract class in JAVA.	L3	8

		Module-5		
Q. 09	a	What is a package ? Explain with example.	L2	6
	b	Demonstrate the working of interface in JAVA.	L2	8
	c	Explain different access modifiers in JAVA.	L3	6
		OR		
Q. 10	a	What is an exception ? Give an example.	L2	6
	b	Explain the procedure of creating our own exception.	L2	8
	c	Explain different mechanism of handling the exceptions in JAVA.	L3	6

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MAKE-UP EXAM

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BPOPS103/203

First/Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023
Principles of Programming using C

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Compare the generation of computers.	10	L2	CO1
	b.	Describe any two input devices.	05	L2	CO1
	c.	Design an algorithm and program to find area of a circle.	05	L3	CO2
		OR			
Q.2	a.	Explain the various rules for forming identifier names. Give an example for valid and invalid identifier.	08	L2	CO2
	b.	Compare between primary memory and secondary memory along with examples.	06	L3	CO2
	c.	Design a flowchart to calculate the sum of first 10 natural numbers.	06	L3	CO2
		Module – 2			
Q.3	a.	Develop a C program that takes 3 coefficients (a, b and c) of a quadratic equation $ax^2 + bx + c$ as I/P and compute all the possible roots and print then with appropriate messages.	08	L3	CO5
	b.	Distinguish between the break and continue statement.	06	L2	CO2
	c.	Describe any 4 types of operators in C with example.	06	L2	CO2
		OR			
Q.4	a.	Develop a simple calculator program in C language to do simple operations like addition, subtraction, multiplication and division. Use switch statements in your program.	08	L3	CO2
	b.	Comparison between typecasting and type conversion with an example.	06	L2	CO2
	c.	Explain while loop along with syntax and example program.	06	L2	CO2
		Module – 3			
Q.5	a.	Implement matrix multiplication and validate the rules of multiplication.	10	L3	CO3
	b.	Summarize the recursive function concept with suitable example.	05	L2	CO3
	c.	Explain declaration and initialization of 1D and 2D Array with an example for each.	05	L2	CO3
		OR			
Q.6	a.	Explain the syntax of function declaration, function call and function definition with an example C program.	06	L2	CO4
	b.	Describe the different types of storage classes with an example.	08	L2	CO3
	c.	Write a C program to sort the array elements in ascending order.	06	L2	CO3
		Module – 4			
Q.7	a.	Develop a C program to compare 2 strings without using built in function.	06	L2	CO4
	b.	Comparison between print( ) and puts( ) functions.	04	L2	CO4
	c.	Define String. Explain any 4 string manipulation function with an example.	10	L2	CO4
		OR			
Q.8	a.	Define a pointer. Summarize the arithmetic operations performed on pointers.	06	L2	CO3
	b.	Develop a C program using pointer to compute the sum, mean and standard deviation of all elements stored in any array of N real numbers.	08	L3	CO3
	c.	Differentiate between NULL pointer and void pointer with suitable example.	06	L2	CO3

Module – 5

Q.9	**a.**	What is a structure? Explain C syntax of structure declaration with an example.	**06**	**L2**	**CO3**
	b.	Implement structures to read, write and compute average marks of the students, list the students scoring above and below the average marks for a class of N students.	**08**	**L3**	**CO5**
	c.	Differentiate structures and unions with syntax and example.	**06**	**L2**	**CO3**
		OR			
Q.10	**a.**	Discuss the different modes of operations on files with suitable example.	**08**	**L3**	**CO4**
	b.	Write a C program to copy a text file to another, read both the input file name and target file name.	**06**	**L3**	**CO4**
	c.	Differentiate between putc( ) and fputc( ) function.	**06**	**L2**	**CO4**

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04-12-2023 09:09:07am
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